Deep inelastic scattering on polarized lithium: the physics case for ^{6}Li and ^{7}Li beams at the Electron-Ion Collider

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Abstract. Polarized ^{6}Li and ^{7}Li beams would give the Electron-Ion Collider (EIC) spin-polarized nuclei that are light enough for ab initio computation yet carry genuine nuclear structure: ^{6}Li is a spin-1 nucleus well described as a deuteron bound to an inert α particle, ^{7}Li its spin-3/2 partner, an α plus a triton. We review why lithium is the practical choice — an alkali atom that optical pumping polarizes, a pair of cluster nuclei whose spin sits on the light cluster, and ions whose spin dynamics in the ring are deuteron-like — and why the physics requires both polarization and a collider. Starting from the spin-1 formalism of inclusive deep inelastic scattering we develop the four measurements the programme is built around: the tensor structure function b_1, the double-helicity-flip structure function Δ (nuclear gluonometry), the polarized EMC effect on the bound proton of ^{7}Li, and spectator-tagged scattering on the embedded deuteron. For each we state what the beam must provide and what one EIC year (10 fb^{-1} per nucleon) reaches. Tensor polarization is the scarce resource, the transverse spin direction the natural one, and the spectator-tagged and coherent channels the ones that fix the far-forward requirements.

1 · Introduction

The EIC project scope provides polarized electrons, protons and ^{3}He; polarized deuterons are a later addition, and no heavier polarized ion appears in the conceptual design or the Yellow Report [1,2]. The EPIOS white paper makes the case for polarized ^{6}Li and ^{7}Li as "a desirable addition", listed with ^{2}H and ^{3}He as critical technology [3], and the ANL source development specifies the beam: an atomic-beam source with optical pumping at 671 nm feeding EBIS charge breeding, with source targets $P_z \geq 0.90$, $P_{zz} \geq 0.80$ and $\geq 10^{13}$ atoms/s [4]. Four measurements motivate the effort, each addressing a gap in the literature: no b_1 prediction or measurement exists for any nucleus beyond the deuteron; no gluon-transversity projection exists for any EIC target; no study of α-tagged deep inelastic scattering (DIS) on lithium exists; and no lithium-beam luminosity or polarization parameter set has been published.

This note is the physics case behind those measurements, written for colleagues joining the programme. Section 2 argues for lithium as an atom, as a nucleus and as a stored ion; Section 3 for polarization and for a collider; Section 4 develops the spin-1 formalism; Section 5 the four observables; Section 6 collects what each requires of the beam and what the programme's simulations project for one EIC year. The projections themselves — rates, bins, estimators and the reconstructed-level results — are the subject of the companion report [5].

2 · Why lithium

2.1 · The atom: optical pumping and a proven source

Lithium is the lightest alkali metal and its atom is hydrogen-like: a single 2s valence electron whose D lines near 671 nm (670.96 and 670.98 nm in vacuum for ^{7}Li) lie within the range of commercial single-frequency diode lasers, and whose ground-state hyperfine splittings — 228.2 MHz for ^{6}Li, 803.5 MHz for ^{7}Li [6] — are small enough for one laser with an electro-optic sideband to address both hyperfine levels. Circularly polarized light pumps the atoms into a single magnetic substate, the hyperfine coupling transfers the electron polarization to the nucleus, and radio-frequency transitions between Zeeman sublevels set any combination of P_z and P_{zz} [4]. Lithium's low ionization energy (5.39 eV [6]) eases ionization, and fully stripped Li^{3+} is within reach of the EBIS being commissioned to charge-breed the polarized $^3He^{++}$ beam, whose design goal is 2×10^{11} ions per pulse at ≥ 70% polarization [7].

Both isotopes are stable and abundant (7.6% ^{6}Li, 92.4% ^{7}Li), and they are the only stable isotopes of one element with J = 1 and J = 3/2: the stable spin-1 nuclei are exactly ^{2}H, ^{6}Li and ^{14}N, and the stable spin-3/2 nuclei below A = 30 are ^{7}Li, ^{9}Be, ^{11}B, ^{21}Ne and ^{23}Na [8]. One source, one laser system and one injector therefore serve a spin-1 and a spin-3/2 programme. EPIOS's list of where an exotic-glue discovery is most likely — "spin-1 ^{6}Li or ^{14}N, or spin-3/2 ^{7}Li or ^{23}Na" [3] — reduces, once optical pumping is required, to lithium and sodium — ^{14}N is not an alkali and has no atomic-beam polarization route, and boron's electronic structure makes its optical pumping a development project of its own [3] — and lithium is the lighter, cluster-structured, ab initio-computable one.

The source physics has a forty-year history. The Heidelberg tandem delivered $^6Li^{3+}$ at vector polarization 0.57–0.65 in 1977 [9]; by 1987 its atomic-beam source reached P_z, $P_{zz} \geq 0.85$ and accelerated ^{7}Li at $P_{zz} = 0.69$ on target [10] — a tensor figure in that paper's convention, not the spin-3/2 alignment used here; the optically pumped Florida State source delivered more than 150 nA of $^6Li^{3+}$ at $P_{yy} = -1.33 \pm 0.02$ with no polarization loss in the linac [11]; and polarized ^{6}Li, ^{7}Li and ^{23}Na beams served two decades of nuclear-reaction studies [12]. New for the EIC are scale — two to three orders of magnitude beyond these sources — and transport: preservation of the polarization through charge breeding and acceleration beyond 100 GeV/u [3,4].

2.2 · The nuclei: the spin sits on a light cluster

The two stable lithium isotopes are the lightest nuclei beyond A = 3 in which a polarized light cluster is bound to an inert core: in ^{6}Li an α particle (J = 0, T = 0) and a deuteron-like cluster bound by 1.47 MeV against α+d breakup, in ^{7}Li an α and a triton bound by 2.47 MeV [8]. Variational Monte Carlo (VMC) with realistic two- and three-nucleon forces makes this quantitative: the α–d overlap of the ^{6}Li ground state integrates to $N_{\alpha d} = 0.86$ (0.846 S-wave, 0.017 D-wave) and the α–t overlap of ^{7}Li to $N_{\alpha t} = 1.00$ [13]; the no-core shell model with continuum gives an asymptotic D/S ratio of −0.027 for the d+α configuration [14]. In a fully polarized ^{6}Li the α contributes nothing and the two valence nucleons carry the spin: six-body VMC gives 1.924 spin-up and 1.076 spin-down protons (and neutrons) in the M = 1 state, an effective polarization $N\uparrow - N\downarrow = 0.85$ per cluster nucleon [13,15]; three-body α+n+p models, in which α+d accounts for 60–70% of the wave function, give above 90% [16]; and E155, which used ^{6}LiD as its deuteron target, adopted 87% for the effective deuteron polarization [17]. The simulation takes the cluster picture literally for the vector polarization: the α–d relative motion and the deuteron inside it each dilute the alignment of a valence nucleon with the ^{6}Li spin by $1 - (3/2)P_D$ — with $P_D^{\alpha d} = 0.087$, chosen so the α–d factor reproduces E155's 0.870, and $P_D^{d} = 0.045$ (AV18-like) — so the polarized proton and neutron carry 0.870 × 0.9325 = 0.811 of it, the same two D-state probabilities behind the tagged α–d observables of §5.4 rather than transcribed constants. The VMC count above is that same quantity obtained ab initio, 0.848 to three figures, and read backwards it puts the α–d dilution at 0.909 rather than at E155's 0.870 — a 6% α–d D state where the adopted wave function carries 8.7% and the overlap quoted above only 2% of $N_{\alpha d}$; 0.81–0.85 is accordingly the band the adopted value is carried with, and what decides between its ends is provenance rather than the 4.5% between them — the cluster product is the wave function the tagged observables already stand on, where the VMC number would enter as a constant no other observable in the chain constrains. The remaining reading — a counting dilution of two polarized nucleons in six, which is a whole-nucleus 1.0 and 23% larger — is not adopted here. ^{6}Li is therefore a polarized deuteron in the field of an α particle, whose tensor observables can be set against the free deuteron's to isolate the medium effect. The only other stable spin-1 ground state, ^{14}N, shares its spin among p-shell nucleons and lies beyond the reach of exact quantum Monte Carlo (A ≤ 12) [18].

In ^{7}Li the spin is carried by the unpaired proton of the triton cluster, the neutrons being paired [19]. VMC gives proton numbers 1.934 up and 1.066 down and neutron numbers 1.981 and 2.019 [13], i.e. $P_p = 0.87$ and $P_n = -0.04$ (0.866 and −0.037 as quoted in the JLab proposal [20]); the shell model already gives $P_p = 13/15$. Polarized ^{7}Li is thus to the proton what polarized ^{3}He is to the neutron, with the property, decisive for the polarized EMC effect, that the proton sits in a nucleus of normal density. Cloët, Bentz and Thomas set the selection criteria — $|P_n| \ll |P_p|$ and light, because g_1^A is carried by the valence nucleons and is suppressed relative to F_2^A by roughly 1/A — and list ^{7}Li among the candidates "where the nuclear polarization is largely dominated by the valence proton" [21]; the gluon-spin study names ^{7}Li "the most promising case for experimental investigation" [22].

Two further properties matter. The cluster structure makes the nuclei computable: Green's function Monte Carlo with AV18+IL7 reproduces both binding energies (−31.93 vs −31.99 MeV for ^{6}Li, −39.15 vs −39.25 MeV for ^{7}Li), the charge radii to within 3% and, with two-body currents, the magnetic moments (0.835 vs 0.822 μ_N; 3.24 vs 3.256 μ_N) [18,19], so the nuclear corrections entering every extraction — effective polarizations, momentum distributions, spectral functions — carry controlled uncertainties rather than model spread. The quadrupole moment of ^{6}Li is the exception: the measured Q = −0.0806 fm², against +0.286 fm² for the free deuteron [8], arises from a delicate cancellation between the deuteron's own moment and the D-wave part of the α–d relative motion [15,23], and theory reproduces only its smallness. That smallness is put to use in §5.2, where the near-sphericity of ^{6}Li becomes a null test for the coherent channel; ^{7}Li, whose p-shell triton orbit gives Q = −4.00 fm² [8], is the deformed counterpart. The α+d versus α+t comparison changes only the cluster.

2.3 · The ion: deuteron-like spin dynamics and deuteron kinematics

In the ring, ^{6}Li is the easy case. Its gyromagnetic anomaly, G ≈ −0.18, is as small as the deuteron's (−0.14), so it crosses only 81 depolarizing resonances on the way to top energy, against 63 for the deuteron and 2457 for ^{3}He [24]; like the deuteron it is handled with a partial solenoid snake and tune jumps, full Siberian snakes being impractically long for so small a G [3,24], so the deuteron techniques [25] carry over. ^{7}Li (G ≈ +1.52) crosses 573 resonances and needs partial snakes with jump quadrupoles [3,24].

The energies follow from the collider, not from the ion. The hadron and electron rings must share a revolution period, so the hadron Lorentz factor is fixed by the circumference and the magnets supply whatever rigidity that γ demands, up to the 917 T·m cap: the ion menu is γ-matched to the proton's, an isolated point at ≈ 41 GeV/u and a continuous band from ≈ 99 GeV/u up to the species' rigidity limit of 275 GeV × Z/A [3,26]. For ^{6}Li that is 40.8 GeV/u and 99.5–137.5 GeV/u, for ^{7}Li 40.8 and 99.5–117.9 GeV/u. Because ^{6}Li has A/Z = 2 it shares the deuteron's rigidity cap, so at the top configuration the two run at the same per-nucleon energy, 137.5 GeV/u, on the same magnets, and e+d and e+^{6}Li can be compared at identical kinematics; at the γ-matched points the deuteron sits 0.45% higher — 41.0 against 40.8, 100 against 99.5 GeV/u — the binding-energy difference per nucleon. The stable spin direction in the arcs is vertical, so transverse polarization is expected to be native at the interaction point — an assumption Report 3 §7 carries with its owner, no IP6 spin configuration having yet been worked out for a lithium fill — while longitudinal polarization needs spin rotators whose settings are equally undefined [1,3].

3 · Why polarization, and why a collider

3.1 · Polarization is the observable

An unpolarized lithium beam would be one more light ion: its inclusive cross section is carried by F_1 and F_2 alone — for an unpolarized target the spin-1 cross section has exactly the spin-½ form [27] — and it would add an A = 6, 7 point to the EMC and shadowing curves that ^{4}He and ^{12}C already map. That point is not without interest: the high-x EMC ratios of ^{6}Li and ^{7}Li are in the target list of JLab E12-10-008 [28], motivated by the finding that the ^{9}Be EMC effect follows the local density of its α clusters [29], and the α-tagged structure function of the triton or deuteron cluster versus spectator momentum would measure the α–t and α–d momentum distributions VMC predicts [13]. But every flagship item of the programme is a spin observable, and the two tensor ones need the tensor moment specifically: b_1 is *defined* as the difference between the m = 0 and m = ±1 quark distributions, so averaging over m erases it, and Δ needs both a target spin of at least 1 and a transverse alignment axis to define an azimuth (§4). The polarized EMC effect is a statement about g_1, and tagged DIS reads the spin of the cluster (§5). Polarized lithium would be the first polarized ion at a collider with a nuclear core — the first place where the spin structure of a *nucleus*, rather than of a nucleon, is studied at collider kinematics — and tensor polarization, on which both b_1 and Δ depend, is the scarcer resource: its survival through charge breeding and acceleration is unmeasured, and its polarimetry requires dedicated development [3,4].

3.2 · Fixed target or collider

Polarized lithium has existed for decades as a fixed target, but only in the solid state: dynamically polarized ^{6}LiD was the deuteron target of SLAC E155 and of COMPASS, ^{7}LiH has been polarized at Saclay (47% for ^{7}Li, 56% for the protons) [30], and ^{7}LiD is the material of the approved CLAS12 polarized-EMC experiment [17,20,31]. In these compounds the lithium polarizes along with the hydrogen isotope because all spin species share one spin temperature: COMPASS reached deuteron polarizations of +54%/−47%, whence 51 ± 3% for ^{6}Li and 92 ± 4% for the ^{7}Li admixture [31]; E155 measured ^{6}Li at 97% of the deuteron's [17]; in ^{7}LiD at 5 T the CLAS12 proposal expects 65–80% for ^{7}Li [20]. Three limits are structural. (i) *Dilution*: only f ≈ 0.35–0.45 of the nucleons in ^{6}LiD are polarizable once the helium coolant is counted — better than 0.22 for ND_3 and 0.15 for NH_3, far from the 1.0 of a stored beam [17,31]. (ii) *Tensor polarization*: in thermal equilibrium it follows from the vector one, $P_{zz} = 2 - \sqrt{(4 - 3P_z^2)}$, so $P_z = 0.5$ gives $P_{zz} = 0.20$; radio-frequency "hole burning" reaches 29–37% in deuterated butanol at the cost of the vector polarization, and the JLab b_1 experiment is conditionally approved on demonstrating 30% [32]. An atomic-beam source prepares the tensor moment directly: HERMES ran its gas target at P_{zz} = +0.89 and −1.65, "a combination which is not possible for solid-state polarized targets" [33]. (iii) *Spectator loss*: a spectator α of 100–300 MeV/c carries only 1.3–12 MeV in the target rest frame and stops within 10–250 μm of lithium deuteride [34], inside cells 2–5 cm long at JLab and 30–60 cm at COMPASS, and a coherent ^{6}Li recoil at $|t| = 0.05$ GeV^2 has only 4.5 MeV. The detectors that do tag slow spectators, BoNuS (protons from 70 MeV/c) and ALERT (α from 230 MeV/c), work with unpolarized gas at a few atmospheres [35,36]; polarized lithium outside dynamically polarized solids has never been used in lepton scattering.

The collider removes each limit at once. The beam is pure lithium, its polarization prepared atom by atom and measured in the ring; the spectator α or triton carries the beam momentum per nucleon into the far-forward spectrometers at a few milliradians; the intact coherent recoil can be tagged down to $p_T \approx 0$ at the secondary focus of a second interaction region [37]; and $\sqrt{s_{NN}}$ up to 99.5 GeV for e(18 GeV) × ^{6}Li(137.5 GeV/u) opens x down to ~10^{-4} at $Q^2 \geq 1$ GeV^2, where JLab at 11 GeV stops at x ≈ 0.06 and HERMES stopped at 0.01 [20,33] — the region where gluons dominate, where HERMES's b_1 was largest (§5.1), and where coherence holds. What stays with the fixed target is luminosity: 10^{35} cm^{-2} s^{-1} (CLAS12 at 10–15 nA, Hall C at 115 nA) [20,32] against an electron–nucleon luminosity of 10^{33}–10^{34} cm^{-2} s^{-1} at the EIC [1,2]. The high-x polarized EMC effect and the high-x b_1 of the deuteron are therefore rightly JLab measurements; the EIC takes the low-x and high-Q^2 regions, the tensor observables at full polarization, and everything that needs a detected spectator or recoil (Table 1).

MEASUREMENT	AT A FIXED TARGET	WHAT ONLY THE COLLIDER ADDS
Polarized EMC effect, ^{7}Li (g_1^A)	Approved: JLab E12-14-001 (CLAS12, polarized ^{7}LiD, 0.06 < x < 0.8, 1 < Q^2 < 15 GeV^2, P_z ≈ 0.8 assumed) [20]	x < 0.06; the Q^2 evolution that carries the gluon-spin EMC effect [22]; an undiluted beam; α-tagging of the struck triton and its virtuality
Tensor b_1, ^{6}Li	Deuteron only: E12-13-011 (ND_3, 0.16 < x < 0.49, P_{zz} ≥ 0.3 required) [32]; ^{6}LiD would give P_{zz} ≈ 0.2 with the deuteron term to be subtracted, and no such measurement is proposed	x down to a few 10^{-3}, below the x ≈ 0.01 where HERMES found its largest b_1 (§5.1); pure-tensor fills at P_{zz} ≈ 0.6–0.8
Exotic glue Δ, ^{6}Li (^{7}Li)	Letter of intent only: LOI12-16-006 (^{14}N in NH_3, x < 0.3, 2–12% alignment, in-plane spectrometers — cos 2φ "difficult with a fixed target") [38]	Transverse tensor polarization ≥ 0.6 with full 2π azimuthal acceptance; the gluon-dominated region x ≲ 0.1 and a Q^2 lever arm
Spectator-tagged DIS (α, t)	Not feasible (§3.2, iii)	Far-forward detection of α and t at beam velocity; pole extrapolation to the free cluster [39,40]
Coherent intact-^{6}Li channel	Not feasible: needs x ≲ 0.01 and a 2–9 MeV recoil	A near-beam tag of the intact recoil (§5.2)

Table 1 — what a fixed target covers and what only the collider adds, measurement by measurement.

4 · Formalism

4.1 · Deep inelastic scattering

An electron with four-momentum k scatters from a nucleus of momentum P by exchanging a virtual photon q = k − k′. The kinematic invariants, in the per-nucleon convention used throughout the programme (energies and structure functions per nucleon, $x \le 1$), are

$$Q^2 = -q^2, \qquad x = \frac{Q^2}{2P\cdot q}, \qquad y = \frac{P\cdot q}{P\cdot k}, \qquad Q^2 = xys, \qquad W^2 = M^2 + Q^2\,\frac{1-x}{x}$$

where x is the momentum fraction of the struck parton, y the energy-transfer fraction and W the invariant mass of the hadronic final state. The unpolarized cross section is carried by two structure functions,

$$\frac{d^2\sigma}{dx\,dQ^2} = \frac{4\pi\alpha^2}{xQ^4}\Big[\Big(1-y+\frac{y^2}{2}\Big)F_2(x,Q^2) - \frac{y^2}{2}\,F_L(x,Q^2)\Big], \qquad F_1 = \frac{F_2}{2x(1+R)}$$

with $R = \sigma_L/\sigma_T$, for which the programme's projections use a placeholder of the right magnitude (0.14–0.18 at the kinematics of interest) and carry the SLAC/E143 R1998 world fit [41] behind a switch whose effect is quantified in §5.2. In the parton model $F_2 = \Sigma_q\, e^2_q\, x[q(x)+\bar{q}(x)]$: the cross section measures quark distributions weighted by squared charge. For a nucleus we use F_2^A/A, so rates are quoted against luminosity per nucleon. Polarization introduces additional structure functions whose number and character depend on the target spin.

4.2 · Spin-1 targets

A spin-½ target contributes two spin structure functions, g_1 and g_2. The principal observable is g_1, the helicity-weighted quark distribution, measured through the double-spin asymmetry $A_\parallel = D\cdot g_1/F_1 \cdot (1 + \gamma^2)$ in the small-y limit of this programme (D is a kinematic depolarization factor and $\gamma^2 = 4M^2x^2/Q^2$ the target-mass term, ≤ 0.0102 rate-weighted in the window of Table 3; the generator has carried it by default since 2026-08-29) with both beams longitudinally polarized and the electron helicity reversed bunch by bunch.

A spin-1 target carries additional structure. Its spin state along a chosen axis is specified by three populations p_+, p_0, p_- of the projections m = +1, 0, −1, and two moments characterize the fill:

$$P_z = p_+ - p_- \in [-1, 1], \qquad P_{zz} = p_+ + p_- - 2p_0 = 1 - 3p_0 \in [-2, 1]$$

P_z is the familiar vector polarization. P_{zz}, the *tensor* or alignment polarization, is specific to spin ≥ 1: it distinguishes spin along ±z from spin in the transverse plane (m = 0) and is nonzero even when equal numbers point up and down. An unpolarized beam has (⅓, ⅓, ⅓) and both moments vanish (Figure 1, left); a fill with all population in m = ±1 reaches P_{zz} = +1 (Figure 1, right), and a pure m = 0 beam the opposite extreme, $P_z = 0$ with $P_{zz} = -2$.

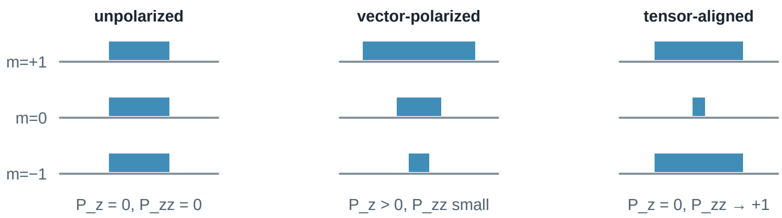

Figure 1 — populations of the three spin projections for an unpolarized, a vector-polarized and a tensor-aligned spin-1 fill. The tensor moment measures the m = ±1 population against m = 0 and is independent of the vector one.

Hoodbhoy, Jaffe and Manohar showed [27] that a spin-1 target has, beyond F_1, F_2, g_1, g_2, four tensor-polarized structure functions b_1–b_4, and Jaffe and Manohar identified one more, the double-helicity-flip function Δ [42]. At leading twist the tensor sector reduces to b_1 (with $b_2 = 2x\, b_1$ as its Callan–Gross partner) and Δ. For unpolarized electrons on a target whose spin axis makes angle θ_m with the beam — which coincides with the virtual-photon direction the exact formula refers to up to O(γ²) target-mass corrections (Report 1 [5] §2) — the cross section for each pure m state is [27,42]

$$\frac{d\sigma^{(m)}}{dx\,dy\,d\phi} = \frac{2y\alpha^2}{Q^2}\Big[F_1 + \frac{2}{3}a_m b_1 + \frac{1-y}{xy^2}\Big(F_2 + \frac{2}{3}a_m b_2\Big) - \frac{1-y}{y^2}\,c_m \sin^2\theta_m\,\Delta(x,Q^2)\cos 2\phi\Big]$$

$$a_m = \frac{1}{4}\,c_m(3\cos^2\theta_m - 1), \qquad c_m = 3|\lambda_m| - 2 \;\rightarrow\; (1,\,-2,\,1) \text{ for } m = (+1,\,0,\,-1)$$

where φ is the azimuth of the lepton plane about the photon direction, measured from the transverse projection of the spin axis. The b_1, b_2 terms are written above in the sign of this programme's own transcription of [27], the form the companion projections [5] display as well; HERMES and Cosyn et al. [43], applying the convention of [27], use the opposite one, A_{zz}

$= -(2/3)\, b_1/F_1$ for the axis along the photon with $P_{zz} = n_+ + n_- - 2n_0$, and since 2026-08-29 so does this programme's simulation — a single constant, adopted because it is the convention every published b_1 is quoted in, so that an extracted b_1 can be compared with them. Only the sign of A_{zz} relative to b_1 is affected here; no magnitude, error or Δ in this report moves with it. Exactly, at every y and free of F_2, the relation reads $A_{zz}(\theta_m = 0)\cdot(1 + \varepsilon(y)R) = -(2/3)\, b_1/F_1$ with $\varepsilon = (1-y)/(1-y+y^2/2)$; the widely quoted form that *also* divides by $[1 + 2(1-y)/y^2]$ and multiplies by $[1 + 2(1-y)(1+R)/y^2]$ counts R twice — those brackets are identically $1 + \varepsilon R$ — and misses by 17%.

The weights (1, −2, 1) organize all tensor observables: the m = ±1 states enter together and against m = 0, precisely the combination measured by P_{zz}, and averaging over a fill replaces every c_m by P_{zz}. Two handles follow. With the spin axis *longitudinal*, the rate difference between fills gives $A_{zz} \propto b_1$; with it *transverse* ($\theta_m = 90°$), the φ distribution of the scattered electron acquires a cos 2φ modulation proportional to Δ. Both require tensor polarization; neither requires polarized electrons.

^{7}Li, with spin 3/2, has four populations and three moments (vector, tensor/quadrupole, octupole). Its rank-0 and rank-1 sectors are exact analogues of the spin-½ case and host the polarized EMC measurement; its rank-2 sector is the formal analogue of the spin-1 tensor sector, for which no structure-function basis has been published. That basis, and the five further theory inputs a ^{7}Li programme waits on, are set out with their asks and their interim assumptions in `docs/note_7li_theory_questions.md`, alongside the two ^{6}Li asks that travel with them.

5 · The observables

5.1 · The tensor structure function b_1

In parton language b_1 compares quark distributions in an aligned versus a transverse nucleus [27],

$$b_1(x) = \frac{1}{2}\sum_q e_q^2\left[q^{(0)}(x) - \frac{q^{(+1)}(x) + q^{(-1)}(x)}{2}\right], \qquad \int_0^1 b_1(x)\,dx = 0 \;\; \text{(Close – Kumano, no tensor sea)}$$

where $q^{(m)}$ is the quark distribution in a target with spin projection m. For free spin-½ nucleons, or a pure S-wave bound state, all m states are equivalent and $b_1 = 0$, so a nonzero b_1 measures directly how binding modifies the partonic content: D-wave admixture, pion exchange, hidden-colour components, or a tensor-polarized sea, which would violate the Close–Kumano sum rule [44].

The only existing measurement is HERMES on the deuteron [33]: A_{zz} up to 2×10^{-2} over $0.01 < x < 0.45$ and $b_1(x = 0.012) = 0.112 \pm 0.055 \pm 0.028$, unexpectedly large at low x, with a zero crossing near $x \approx 0.2$ and a sum-rule integral 2σ from zero. Convolution calculations predict $|b_1| < 10^{-3}$ at $x \gtrsim 0.2$, an order of magnitude below the data [45], whereas a model with pions and a hidden-colour probability of only 0.15% reproduces it [46]. Both curves are digitized into this programme's fast simulation, from Fig. 4 of [45] and Fig. 5 of [46]; over the x range the figures cover, the convolution respects the Close–Kumano sum rule, $\int b_1\, dx = 4.6\times10^{-4}$, and the pionic model does not, 5.9×10^{-3}, as its tensor-polarized sea requires. JLab E12-13-011 will remeasure $0.16 < x < 0.49$ with a solid ND_3 target [32]; the EIC adds the low-x region and the Q^2 lever arm. The beam is delivered in three fills of different tensor polarization; with the estimator $(N_+ + N_- - 2N_0)$ over matched luminosities all m-independent physics cancels [5]. At collider kinematics a clean extraction favours orienting the spin axis along the momentum transfer **q** rather than along the beam [43].

5.2 · $\Delta(x,Q^2)$: exotic glue

The cos 2φ term corresponds to the virtual photon flipping its helicity by two units between absorption and emission in the forward Compton amplitude. A spin-½ particle cannot absorb two units of helicity, so neither quarks nor bound nucleons generate Δ at leading twist; gluons can. Δ is what the gluon transversity distribution — an interference of gluon helicity states that exists only for J ≥ 1 — generates at order α_s through a photon–gluon convolution [42,47], so a nonzero Δ in a nucleus is direct evidence for glue not inherited from individual nucleons, hence "exotic glue". It has never been measured; the one prior experimental document is a letter of intent (Table 1).

The measurement uses unpolarized electrons on a tensor-polarized nucleus whose alignment axis lies in the transverse plane, and fits the azimuthal distribution about that axis. Averaged over the fill, the amplitude and its inversion back to Δ are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2}\frac{\Delta}{D_\phi}, \qquad D_\phi = F_1 + \frac{1-y}{xy^2}F_2$$

Writing $F_2 = 2x(1 + R)F_1$ removes both structure functions from the ratio Δ/F_1 except through R, which is the one structure-function input that reaches the physics result — F_2 cancels exactly. At the four kinematic bins of the companion projection ($x = 0.011$–0.14, $Q^2 = 1.1$–14 GeV^2), replacing the placeholder R by the R1998 fit moves Δ/F_1 by +18, +18, +8 and −4% [5].

The expected size spans per mille to per cent, and the only absolute normalizations available are moment estimates: a bag-model calculation for the spin-3/2 Δ^{++} baryon, $\int x\,\Delta\,dx = -0.012\,\alpha_s$ [47], and lattice-QCD calculations of gluon transversity in the φ meson (clearly nonzero, at an unphysical pion mass) [48], with the analogous matrix element roughly ten times suppressed in the deuteron [49]. Percent-level asymmetries appear only in proton–deuteron Drell–Yan under the maximal assumption $\Delta_T g = \Delta g$, which its authors call optimistic [50]. No calculation exists for ^{6}Li or any A > 2 nucleus, and the literature brackets the nuclear Δ both ways: suppressed, as in the lattice deuteron, or enhanced by binding in analogy with the EMC effect [3]. The projection of [5] transfers the bag moment to ^{6}Li as a scenario — a per-nucleon Δ whose x-moment is one third of the bag value, the ⅓ the counting fraction of the polarized pair among the six nucleons — and one EIC year then measures the amplitude at 21–43σ per bin (Table 3).

The coherent channel. The same modulation can be measured in coherent diffraction, e ^{6}Li → e′ X ^{6}Li(g.s.), where the nucleus stays in its ground state and recoils intact. There the modulation mixes two mechanisms: the nuclear *shape*, a deformed gluon density modulating the diffractive slope (computed for polarized deuterons in the only existing calculation of this kind [51]), and exotic glue proper, which persists at small x, where it connects to the elliptic-gluon programme [52]. ^{6}Li discriminates between them because its quadrupole moment is anomalously small (§2.2): the shape term is bounded and linear in t, so a flat-in-t modulation cannot be a shape effect. Detecting the intact recoil is the difficulty. The nucleus keeps the beam's rigidity and emerges within a milliradian or two of the beam, so the only handle is its angle against the 10σ beam envelope and the Roman-Pot aperture, and the tagged fraction is an exponential in the square of the beam divergence, $\exp[-B(10\sigma_\theta \cdot A \cdot p_u)^2]$, with $B \approx 50$ GeV^{-2} the slope of the coherent t distribution, σ_θ the beam angular divergence and p_u the momentum per nucleon. With the Yellow Report's divergences the fraction is 5×10^{-7}, 8×10^{-26} and 4×10^{-13} at the three ^{6}Li configurations for a circular envelope, and the real anisotropic rectangle with the measured pot edge (horizontal 2.50 / 1.51 / 0.53 mrad against vertical shut / 2.12 / 0.92 mrad, re-measured on 2026-08-28 in the current ePIC geometry, the 5 × 41 horizontal figure standing on the ePIC baseline proton lattice, adopted for a ^{6}Li fill as an educated guess and worth ×0.64 in the Z/A = 0.5 alternative) as a second constraint per axis lowers it further, to 9×10^{-10}, 6×10^{-27} and 7×10^{-14} [26]: the channel does not exist at IP6 with any published optics, though which of the two constraints delivers that verdict has changed hands with the re-measurement — the silicon at 5 × 40.8, where 2.50 mrad exceeds a 2.20 mrad envelope, and the beam at 18 × 137.5, where the aperture halved to 0.53 mrad against an envelope of 0.92. It requires either a dedicated tagging optics — β* raised so that the envelope sits at $p_T = 1/\sqrt{B} \approx 0.14$ GeV, where the tagged fraction is 1/e (a factor 15–60 in β* for a circular envelope), with pots that follow — or the secondary focus of the second interaction region, which reaches $p_T \approx 0$ [37]. The tagging optics recovers a 25–37% tag with the horizontal β* raised 46–164× and the vertical plane untouched, at a factor 6.8–12.8 in luminosity, which makes it an 8–11σ-per-year measurement of the shape term (quoted on the anchor's a_2 footing; the observed cos 2φ modulation is $2a_2$, so the significance is conservative by ×2 — Report 1 [5] §6.3) and a three-to-six-year one of a 1% exotic-glue term, both on the counting floor of its best (x, Q^2) super-bin alone (the seven-|t|-bin fit of the whole tagged sample measures the flat term to 0.07–0.12% per unit P_{zz} per year). That is 4–10× below what a second interaction region's secondary focus — ≈ 20% for ^{6}Li, our interpolation between the published d, ^{3}He, ^{4}He and ^{7}Li efficiencies [37], no ^{6}Li case being published — would give at the same luminosity, itself unpublished [5]; the projections of [5] assume a 0.20 GeV envelope, 13.5% of the coherent yield, at the high-acceptance luminosity, as the reference tag.

5.3 · The polarized EMC effect: g_1 of a bound proton

The EMC effect — the ~10–20% suppression of F_2 per nucleon in a nucleus at $0.3 \lesssim x \lesssim 0.7$ — has been measured for forty years without a settled explanation. Its spin analogue has never been measured: does the *spin* structure of a bound nucleon change more, less, or the same as its momentum structure? The observable is the polarized EMC ratio

$$\Delta R_A(x) = \frac{g_1^A(x)}{P_p\, g_1^p(x) + P_n\, g_1^n(x)}$$

where P_p, P_n are the effective nucleon polarizations from nuclear structure. With the effective polarizations of §2.2, ^{7}Li is to a good approximation one polarized proton plus six spectators, with computable corrections. Two classes of model differ sharply: Cloët, Bentz and Thomas (mean field, NJL) obtain a polarized EMC effect about twice the unpolarized one [21], Tronchin, Matevosyan and Thomas equal effects [53]. Both calculations are digitized here from their published figures, which sharpens the comparison and moves it. Within Cloët, Bentz and Thomas's own ^{7}Li calculation — their Eq. (23) $R^{3/2,3/2}$, the curve plans/01 defines ΔR_A on, with their Eq. (26) $R^{(3/2,1)}$ digitized alongside it — the ratio of effects $(1 - R_{pol})/(1 - R_{unpol})$ is 2.25, 1.69, 1.41 and 1.14 at x = 0.40, 0.45, 0.50 and 0.60, falling to a minimum of 1.06 at x ≈ 0.70 without reaching unity, against 1.01, 0.98, 1.00 and 1.08 for Tronchin, Matevosyan and Thomas: the factor two is a statement about x ≈ 0.4 rather than a constant, and the two converge at the top of the valence region instead of separating there. The two are computed for different targets and scales — ^{7}Li at Q^2 = 5 GeV² and isospin-symmetric nuclear matter at Q^2 = 10 GeV² — so a projection needs them on one common unpolarized baseline, matched here in the strength of the unpolarized effect over 0.35 < x < 0.65. Since 2026-08-29 that baseline is measured rather than modelled — EPPS21's ^{6}Li F_2 per nucleon over CT18ANLO's free isoscalar nucleon — EPPS21's own proton baseline, so the proton fit cancels and what is left is the nuclear modification — whose mean valence depletion over the window is 0.0311, and both camps are rescaled onto it, by 0.53 and 0.21 where the mean-field calculation's own ^{7}Li curve (depletion 0.0584) gave 1 and 0.40. Carried across that way their predictions for ΔR differ by 0.023, 0.021, 0.018 and 0.032 at x = 0.09, 0.28, 0.45 and 0.71 — half of what the model baseline gave, and the choice of baseline, not the statistics, is now the leading uncertainty on the comparison: nNNPDF3.0 reads 0.0137 over the same window, and EPPS21's own 90% CL Hessian band on it is +0.039 / −0.041. That transfer factor is a single constant fitted in the valence window and then applied at every x, so only the part of the separation that falls inside the window is a statement about the two calculations. Below it they agree: the two published polarized ratios differ by less than 0.008 across the whole decade 0.028 < x < 0.30 and by 0.002 at x = 0.09, and ^{7}Li's unpolarized effect there is an enhancement rather than a depletion, so a strength ratio has nothing to scale — the near-constant 0.02 the projection plots at low x is the rescaling, not the papers. Inside the window the comparison is the one the two camps actually make, and it fades across it: the transferred nuclear-matter depletion tracks the baseline's own 0.021, 0.027 and 0.041 at x = 0.40, 0.45 and 0.65 to within 0.003, which is what "polarized ≈ unpolarized" means, against 0.041, 0.043 and 0.050 for the mean-field calculation, so ΔR separates by 0.021 at x = 0.36 and by only 0.006 at 0.65 as the ratio of effects falls to unity. The discriminating region is therefore x ≈ 0.35–0.45 — inside the valence region, as before the digitization, but at its lower edge and with a quarter of the separation the constants 2 and 1 implied. A second effect points the same way: the three-flavour g_1 over five-flavour F_2 scheme of the two grids Table 3 names, NNPDFpol11 setting Δc = Δb = 0 where CT18NLO carries charm and bottom, costs 0.3% of F_2^A at x = 0.35 falling to 0.03% at 0.65, against 2.6% at x = 0.09, where it is as large as the separation drawn. The two map onto the competing mean-field and short-range-correlation pictures of the EMC effect itself. Unique to the EIC is the Q^2 evolution of g_1, driven by the polarized gluon distribution; the predicted *gluon-spin* EMC effect is larger still, with ^{7}Li the most promising case [22], and the collider measures it free of target-fragment absorption. The measurement uses the longitudinal double-spin configuration of §4.2, i.e. the vector polarization the source delivers most readily. ^{6}Li and the deuteron supply the isoscalar companions, so the three together separate the u- and d-quark spin modification — an inference of this programme, not yet in the literature.

5.4 · Spectator-tagged DIS

Detecting the α spectator, which carries beam velocity into the far-forward spectrometers, identifies the struck object as the deuteron cluster of ^{6}Li or the triton cluster of ^{7}Li, so the beam acts as a source of polarized light clusters at controlled virtuality. The framework is spectator tagging [39]: the spectator momentum is measured and extrapolated in its transverse component to the free-cluster pole, recovering free-particle structure with binding effects removed by construction rather than modelled. Tagged *tensor* asymmetries reach O(1) at spectator momenta of ~300 MeV/c, where the D-wave dominates [54], far larger than the inclusive asymmetries. Cosyn and Weiss give that asymmetry in closed form and this programme's generator reproduces it: at fixed spectator momentum the wave-function tensor asymmetry factorizes into their $P_2(\cos\theta_k)$ angular factor to better than 10^{-5} in every angular cell away from the zero of P_2 at $\cos\theta_k = 1/\sqrt{3}$, where the ratio is unbounded, its radial envelope peaks at k = 0.310 GeV/c against their 0.30 GeV/c for the AV18 wave function, and under the normalization $A_{T\parallel} = -2\, A_{zz}^{wf}$ its extremes extrapolate to +1.000 at $\theta_k = 90°$ and −2.000 at $\theta_k = 0$, the two values they tabulate. The benchmark is the e+³He double-tagging study of Friščić et al. [55], where tagging improved on effective-polarization unfolding by a factor 2–3, and 10 at low x.

Where each fragment is detected follows from magnetic rigidity alone: a beam-velocity fragment bends in the hadron optics according to the ratio of mass-to-charge ratios $R = (m/Z)_{frag}/(m/Z)_{beam}$ (Table 2). The (A/Z) ratio approximates it to 5×10^{-3}; the α is more bound per nucleon than the lithium it leaves, so it sits just below beam rigidity rather than on it.

BEAM	TAG FRAGMENT	R	DESTINATION AT IP6
⁷Li (A/Z = 7/3)	α	0.85571	Roman-Pot main window (R ∈ 0.60–0.95) — works now
⁷Li	t	1.28971	over-rigid: Roman-Pot inner side + ZDC (measured; see text)
⁶Li (A/Z = 2)	α / d	0.99813 / 1.00452	within 5×10^{-3} of beam rigidity: inside the beam envelope except the p_T tail
⁶Li	p / n	0.50251 / –	Off-Momentum Detectors / ZDC (veto handles)

Table 2 — far-forward routing of the tag fragments at IP6, from the rigidity ratio R alone; the windows are those of Jentsch, Tu and Weiss [40]. The over-rigid triton was counted as lost until 2026-08-28, on the argument that the rigidity windows carry no R > 1 branch; that is now settled by measurement rather than assumption. A full simulation of the current ePIC geometry puts a triton on the Roman-Pot silicon in 60 of 60 events at every configuration — the gun was fired at R = 1.286 against the physical-mass 1.28971 of the table, a 0.3% rigidity difference worth under 0.3 mm at the pot plane —, on the inner side of the bend at dx = +66 mm, with a zero-degree-calorimeter deposit behind it in 80–98% of the same events, so the ⁷Li α + t double tag is available at IP6 as an RP-inner track plus calorimeter energy [26]. What remains unchecked is a reconstruction: the hit is at y ≈ 0 and needs no p_T, but the fast simulation routes in angle and cannot see the per-band vertical insertions that put a real acceptance hole between θ_x = −1.55 and −2.53 mrad at 18 × 117.9.

In the cluster model of [26], the ⁷Li α tag reaches the Roman Pots with ≈ 97% acceptance at every configuration and every optics (95–99% over the wave-function tail scale β = 0.2–0.4 GeV), because R = 0.856 puts it in the main Roman-Pot window and no beam envelope is in its way. That routing is in angle alone, and at 5 × 40.8 it is optimistic: the per-energy insertion holds the silicon band off to |y| ≥ 18.0 mm there, so an α at y ≈ 0 sees no silicon at station 1 at all and the full simulation of the current geometry finds it at station 2 only, on a module edge, in 57% of events (Report 3 [26] Table 6). At 10 × 99.5 and 18 × 117.9 the insertion is nearer and the angle-only 97% stands. The ⁶Li α is the opposite case. It sits at R = 0.998, so it can be tagged only through the p_T tail above the near-beam cut, and whether that tail exists is an optics question. At the Yellow Report high-acceptance optics it barely does: almost all of the tail lies inside the 10σ envelope, and the 1.9 / 1.7 / 2.6% the tag reads at 40.8 / 99.5 / 137.5 GeV/u is dominated by the off-rigidity slice the Roman-Pot window catches together with the over-rigid inner branch, with 0.20 / 0.01 / 0.08 points of genuine near-beam tail beside them — a ninth of the tag at 5 × 40.8 and a thirtieth or less at the other two. The tagging optics of §5.2 reopens the tail and recovers 31 / 22 / 29% at 1/6.8–1/12.8 of the luminosity, the same trade and the same request as the coherent channel, and Report 3 [26] Table 6 tabulates both tags against both optics. Those acceptances are the pure cluster-model routing, one partial wave per channel; the tagged tensor asymmetry above rides the α–d D wave, which is hard (⟨k⟩ = 0.28 against 0.11 GeV/c) and populates exactly the off-rigidity slice the pots see, so the S + D generator that produces the observable reads 2.5% at the Yellow Report optics and 25% at the tagging optics at 10 × 99.5 (the 2.5% in Report 3 [26] Table 6, the 25% in Report 4 §2.1). The triton of ⁷Li is the one tag the re-measurement transformed: routed as lost while R = 1.29 had no window, it is tagged in 78 / 92 / 94% once the measured over-rigid branch is applied, almost all of it on the inner side of the Roman Pots and the rest in the B0 at 5 × 40.8, where the angles are largest (Table 2). Because the ⁶Li tag under the tagging optics is a p_T-tail measurement, its model uncertainty there is one-sided upward: the cluster model undershoots the official BeAGLE e+d sample in the spectator peak by 2.5×, 6.9× and 28× at $p_T > 0.2$, 0.3 and 0.45 GeV. The tagging programme therefore begins with ^{7}Li at IP6, while ^{6}Li α-tagging — and the coherent channel, which faces the same rigidity problem — motivates the near-beam tagging capability of §5.2.

6 · Requirements and reach

All projections use luminosity per nucleon: 10 fb^{-1}/u — 30 weeks at 10^{33} cm^{-2} s^{-1} with 60% efficiency in the Yellow Report's accounting [2], "one EIC year", taken as the same placeholder at every configuration, the Yellow Report's own per-configuration luminosities differing by up to 23× (Report 3 [26] Table 3) and no lithium luminosity being published — and 100 fb^{-1}/u for the ten-year programme, at placeholder in-ring polarizations $P_z = 0.7$, $P_{zz} = 0.6$ and $P_e = 0.7$; transport of the source polarization to the ring is an open accelerator question, and every statistical error scales as $1/(P\sqrt{L})$, so halving P_{zz} costs a factor four in running time. The observables do not share a configuration: they demand different spin states, different far-forward optics — the coherent channel of §5.2 wants the tagging optics and pays its luminosity penalty for as long as it runs — and, between the polarized EMC effect and the tensor sector, different isotopes, ^{6}Li and ^{7}Li being separate fills and separate runs, and each projection assumes the full luminosity of the year in its own. The reaches of Table 3 are therefore per observable and cannot be added into one year; how the year is divided between them is a programme decision, priced in plans/07 WP2, under which every statistical error grows as one over the square root of the share while a luminosity quoted as a reach is unchanged by it. The statistical uncertainties behind every projection are

$$\delta A_{\parallel} = \frac{1}{P_e P_z \sqrt{N}}, \qquad \delta A_{zz} = \frac{\sqrt{2/N}}{P_{zz}}, \qquad \delta A^{\cos 2\phi} = \frac{\sqrt{2/N}}{P_{zz}} \text{ (single state)}, \qquad \delta\hat{A} = \frac{2}{P_+ - P_0}\sqrt{\frac{2}{N}} \text{ (two states)}$$

The factor √2 in the tensor uncertainties follows from $\langle\cos^2 2\varphi\rangle = ½$ (equivalently, from the three-fill estimator variance); the polarization sits in the denominator because the fill dilutes the physical asymmetry. These formulas, and their behaviour under realistic φ-acceptance holes (where a binned fit replaces the naive moment), are validated by the pseudo-experiments of [5]. Two of the four highest-significance Δ bins sit at $y \approx 0.01$ and the other two at $y \approx 0.025$, where the electron alone gives $\delta y/y \approx 1.1$–1.2 and 0.46 and the hadronic Σ method, with a PYTHIA 8 final state and 50 MeV of calorimeter noise, 0.32 / 0.22 / 0.29 / 0.15 at the mid configuration — a resolution set by the noise floor (0.2 → 1.0 at y = 0.005 for 0 → 100 MeV) on a hadronic $E - p_z$ sum of 0.2–0.5 GeV that no published ePIC study covers. Tensor observables have a structural advantage when the m = ±1-rich and m = 0-rich states are measured with the same detector and compared bin by bin. In their spin-state ratio the φ acceptance cancels and relative-luminosity errors enter only as a constant, whereas a single-state fit also measures the detector's own cos 2φ modulation divided by P_{zz}. The two-state pattern $P_+ = +0.6$, $P_0 = -1.2$ is moreover 1.5× more precise than a single state, provided the source delivers the m = 0-rich bunches at the same purity. The cancellation requires bunch-by-bunch alternation: a 10^{-3} difference of the acceptance harmonic between the states fakes half a 10^{-3} signal. The polarimetry scale and the tensor radiative corrections do not cancel. Table 3 collects, observable by observable, what the beam must provide and what the simulations project.

OBSERVABLE	BEAM POLARIZATION	ELECTRONS	REACH AT 10 FB⁻¹/U	BINDING REQUIREMENT
g_1^A, polarized EMC (^{7}Li; ^{6}Li for the isoscalar pair)	Vector $P_z \geq 0.7$, **longitudinal** (spin rotators). A spin-3/2 spin-temperature fill at $P_z = 0.8$ also carries $P_{zz} \approx 0.55$ and $P_{zzz} \approx 0.11$ in the convention of [20], which is 0.34 in the rank-3 normalization `polligen.spin` adopts — the identity between the two is open item 1 of `docs/note_7li_theory_questions.md`; in [20] the octupole term enters the asymmetry numerator and is corrected through the fill model	Polarized, $P_e \approx 0.7$, helicity flipped bunch by bunch	δΔR ≈ 4.2% per x-bin at x = 0.09, 4.0% at 0.28, 6.0% at 0.45 and 18.7% at 0.71 (three energies combined, on CT18NLO + NNPDFpol11 grids; 4.8, 5.1, 6.2 and 12.2% on the toy inputs). Against the digitized curves on the EPPS21 baseline (§5.3) the CBT–TMT difference is 0.55, 0.53, 0.31 and 0.17σ per bin there on the grid inputs, and 1.75, 1.66, 0.97 and 0.53σ with 100 fb^{-1}/u — nothing	Polarimetry scale $\delta P_z/P_z \leq 3\%$; P_p from VMC [13]; a Q^2 lever arm at fixed x (gluon-spin EMC); the target-mass term of $A_\parallel$, which the generator has carried by default since 2026-08-29, is worth 0.12 / 0.46 / 0.73 / 1.08% of ΔR at x = 0.09 / 0.28 / 0.45 / 0.71 (inverse-variance weighted over Q^2 and the three energies — the same weights as the reach

			reaches 1σ per bin at 10 fb^{-1}/u anywhere. Read where the two calculations actually differ — the valence window $0.35 < x < 0.65$ in which the transfer of §5.3 is defined — the best bin is $x = 0.355$ at 0.45σ at 10 fb^{-1}/u and 1.43σ at 100 (0.38 and 1.22σ on the toy inputs); the larger figures the same projection returns at $x \approx 0.09$–0.14 are carried by the transfer rather than by the papers	column's toy-input δΔR, 4.8 / 5.1 / 6.2 / 12.2%, the inputs the term is computed on) and is now computed rather than left as a bias; what it leaves as a systematic is the twist-3 uncertainty on g_2, ±0.002 / ±0.034 / ±0.078 / ±0.159% over $g_2 = 0$ to $1.5\ g_2^{WW}$, a factor 77 / 13 / 9.4 / 6.8 under the term in the same four bins and 19–1900× below the polarimetry scale
b_1 via A_{zz} (^{6}Li)	**Tensor** P_{zz}, longitudinal or along **q** [43]; fills of different P_{zz} ("tensor thirds"), i.e. m = 0 bunches from the source	Unpolarized (helicity-averaged)	$\delta A_{zz} \approx 0.9\times10^{-4}$ per x-bin (combined over Q^2 and the three energies) at $x < 0.05$ (0.7×10^{-4} at $P_{zz} = 0.8$), rising to $\approx 10^{-3}$ at $x \approx 0.6$, against the digitized b_1 of the embedded deuteron [46], carried into ^{6}Li by the per-deuteron→per-nucleon ½, the rank-2 tensor transfer 0.92 and the ⅓ dilution — 0.15 in all, or 0.31 applied to a b_1 already quoted per nucleon (not the vector 0.87, which is the wrong rank for a tensor structure function): $\lvert A_{zz}\rvert = 2.2\times10^{-4}$ at $x = 0.005$ rising to 1.4×10^{-3} at $x = 0.07$, and, with the signal interpolated to each error bin's centre, 1.6, 4.7, 7.3 and 10.4σ per bin at $x = 0.0035$, 0.0089, 0.028 and 0.071 at $P_{zz} = 0.6$ — the signal read at each bin's own $\langle Q^2\rangle$ and rate-weighted y, 3.2 to 8.9 GeV^2 over those four, since 2026-08-29 — while the convolution [45] stays below 0.2σ everywhere: the two calculations are not a factor ten apart but the difference between a measurement and none. $x > 0.3$ is JLab's domain	Relative luminosity between fills at the 10^{-4} level (a 10^{-4} offset biases A_{zz} by 1×10^{-4}; the luminosity-corrected estimator removes it exactly); tensor polarimetry, which EPIOS flags as requiring dedicated development (Lamb-shift or Breit–Rabi-type polarimeters) [3]
Δ via cos 2φ (^{6}Li; ^{7}Li eligible)	**Tensor** P_{zz} with the alignment axis **transverse** (the native vertical direction); fills of opposite P_{zz} sign flip the modulation — the decisive systematic check	Unpolarized	$\delta A = (1.4$–$4.5)\times10^{-4}$ per bin in the four highest-significance bins ($x = 0.011$–0.14, $Q^2 = 1.1$–14 GeV^2) against bag-moment-constrained amplitudes of $(0.44$–$0.95)\times10^{-2}$ at $P_{zz} = 0.6$, i.e. 21–43σ; a flat	Spin-state ratio of m = ±1-rich and m = 0-rich bunches alternating bunch by bunch; a hadronic-method y with $\delta y/y \lesssim 0.3$ at the $y \approx 0.01$–0.03 bins, i.e. a calorimeter noise floor ≲ 50 MeV on a hadronic

			amplitude of 3×10^{-3} per unit P_{zz} stays ≥ 3σ per bin even at P_{zz} = 0.3. Reconstructed, with the two-state ratio and a hadronic-method y: $\delta\hat{A}$ = $(0.9–3.0)\times10^{-4}$; with a PYTHIA 8 final state through the hadron-side response and the hadronic scale calibrated, the bin purities are 0.56–0.75 and the reconstructed-bin amplitude within 7% of the true-bin value (0.42–0.68 and 5–21% low uncalibrated)	$E - p_z$ sum of 0.2–0.5 GeV (text); $\delta P_{zz}/P_{zz}$ ≤ 5% scale, the working value this programme specifies (3% is the optimistic case, reached on stored deuterons at the Nuclotron and on the HERMES storage cell), entering Δ as a 1:1 normalisation [5]; tensor radiative corrections not yet calculated
Spectator-tagged DIS	^{7}Li α-tag: any polarization for tagged F_2; vector P_z with polarized electrons for tagged g_1 of the bound proton. ^{6}Li α-tag: tensor P_{zz} for the order-unity tagged tensor asymmetries (§5.4)	As for the inclusive observable	^{7}Li α → Roman Pots, ≈ 97%, optics-robust in angle (at 5 × 40.8 the measured per-band insertion leaves 57%, station 2 only); ^{7}Li t → Roman-Pot inner side + ZDC, 78–94%; ^{6}Li α beam-blind at IP6, 1.7–2.6% at the Yellow Report optics of the three configurations and 22–31% under the tagging optics of §5.2, at 1/6.8–1/12.8 of the luminosity (§5.4)	Spectator p_T resolution ~10% at 100 MeV/c [40]; charge identification of α against A/Z = 2 fragments in the Roman Pots; pole extrapolation in p_T [39]; the one-sided p_T-tail model uncertainty
Coherent intact-^{6}Li cos 2φ	**Tensor** P_{zz} ≥ 0.6, **transverse**	Unpolarized	x ≤ 0.01, |t| in the published seven-bin analysis window 0.017–0.25 GeV² [5]. For a tag with a 0.20 GeV near-beam envelope (13.5% of the coherent yield, i.e. |t| > 0.04 GeV² with ⟨|t|⟩ = $p_{T,cut}^2 + 1/B$ = 0.06 GeV²): 1.7×10^7 tagged events, best-bin δA = 1.8×10^{-3} (5σ at a 0.9% modulation per unit P_{zz}), against a shape term ⟨a_2⟩ = 0.060 [0.030–0.098] per unit P_{zz} (0.036 [0.018–0.059] at P_{zz} = 0.6) scaled from the polarized-deuteron calculation [51] and a flat gluonic 3×10^{-3}–10^{-2} on the same per-unit footing [5]	A near-beam tag (§5.2); rejection of α + d breakup, both at beam rigidity, by the |t| shape and the partner-fragment hit — pot charge identification would add to it; B0 calorimeter veto of the 3.56 MeV de-excitation γ

Table 3 — beam requirements and one-year reach by observable, from the programme's fast simulation and event-generator pseudo-experiments. The Δ and coherent rows follow [5] and are at e(10 GeV) × ^{6}Li(99.5 GeV/u); the b_1 and polarized-EMC rows, published here rather than in a companion, combine the three beam configurations; the spectator-tagged row is drawn from Report 3 [26] Table 6, where the ^{6}Li tagging acceptances span the three ^{6}Li configurations (40.8 / 99.5 / 137.5 GeV/u), and are the S-wave cluster-model routing — the tensor-tagged channel is quoted on the S + D spectrum, 2.5% and 25% at 10 × 99.5 (§5.4); the ^{7}Li rows are the three ^{7}Li configurations (40.8 / 99.5 / 117.9 GeV/u, the top one rigidity-capped). The signal

structure functions — b_1, Δ, the polarized-EMC ratios and the ^{6}Li transfers — are scenarios with explicit bands; the unpolarized backbone they ride on is fitted rather than assumed (CT18NLO and NNPDFpol11 for the rates and δΔR, EPPS21's ^{6}Li over CT18ANLO for the baseline of §5.3, whose fit-to-fit spread is the leading uncertainty on the g_1^A row).

Species by species: ^{6}Li is the only spin-1 nucleus beyond the deuteron the EIC could polarize — b_1 and Δ do not exist for protons or ^{3}He at all — and the lattice indication that the deuteron's exotic glue is suppressed [49] makes a bound nucleus the likelier place for a discovery [3]. ^{7}Li is the only bound polarized proton within reach of any collider, and the only light-ion beam whose spectator is a spin-0, tightly bound nucleus landing in the Roman-Pot window today. Figures 2 and 3 show the phase space and the projected Δ measurement that Table 3 summarizes.

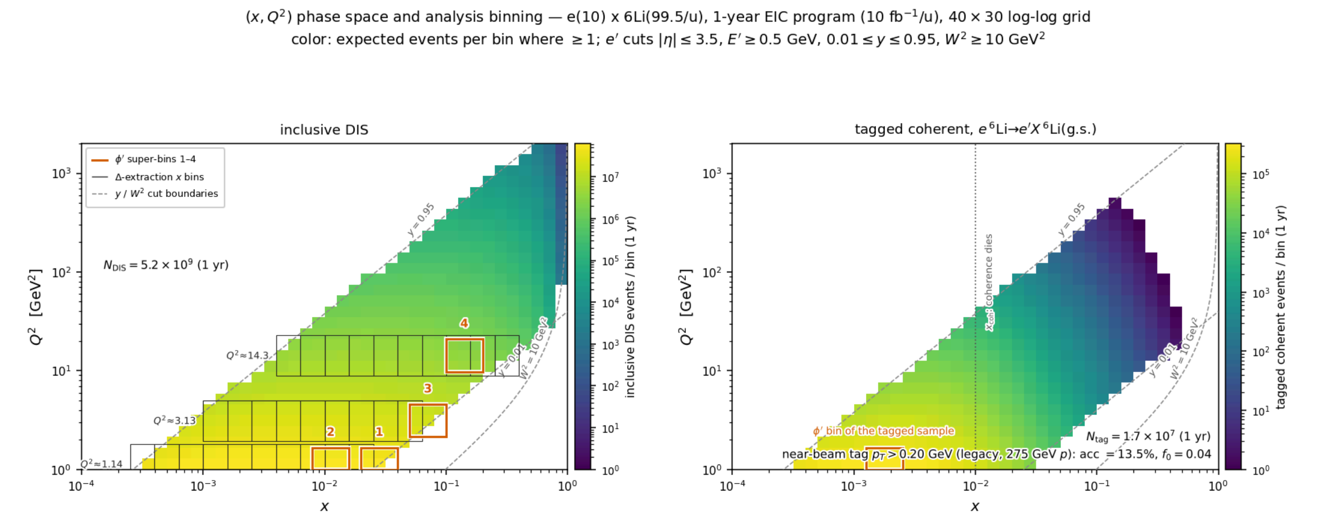

Figure 2 — expected events per (x, Q^2) bin in one EIC year at e(10 GeV) × ^{6}Li(99.5 GeV/u). Left: inclusive DIS (5.2×10^9 accepted events) with the bins of the cos 2φ analysis overlaid; the measurement is limited by polarization, acceptance and systematics, not by statistics. Right: the coherent intact-^{6}Li channel for a tag with a 0.20 GeV near-beam envelope (1.7×10^7 tagged events; the panel's "legacy, 275 GeV p" label is the proton optics the cut was first taken from — no published ^{6}Li optics provides it, §5.2), confined to x ≲ 0.03: the coherent fraction halves at the scale x_{coh} = 0.01 Tables 1 and 3 quote — diffractive practice x_P < 0.01 — the coherence length ≈ 0.1 fm/x reaching the nuclear diameter (≈ 5 fm) only at x ≈ 0.02.

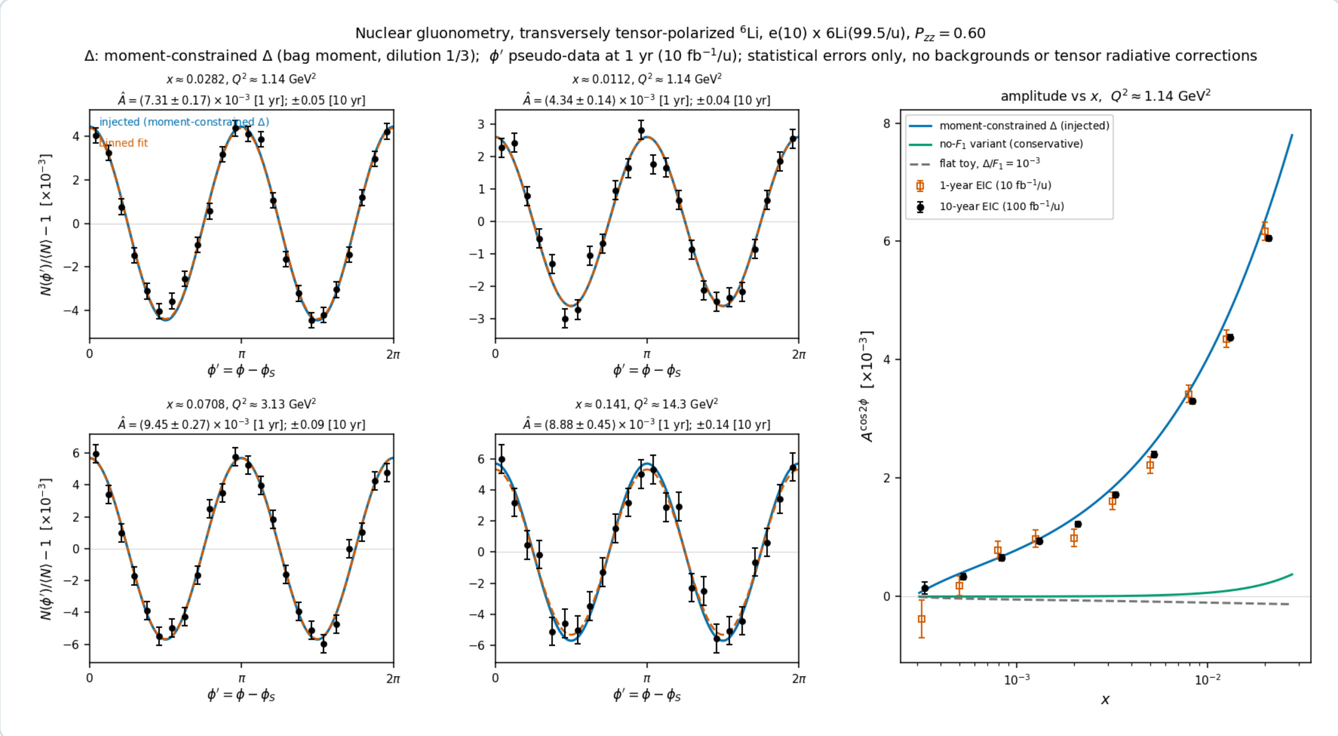

Figure 3 — projected extraction of Δ. cos 2φ pseudo-data (one EIC year, P_{zz} = 0.6) in the four highest-significance (x, Q^2) bins, with the injected moment-constrained Δ model and the binned fit; right, the extracted amplitude versus x at $Q^2 \approx 1.14$ GeV^2, one-year (open) and ten-year (filled), against alternative model readings. The panels' abscissa φ′ is the φ of §4.2, written on the axis as the lab azimuth of the scattered electron minus that of the spin axis. The precision distinguishes between the models rather than only from zero. The reconstructed-level version uses the spin-state ratio in reconstructed bins [5].

7 · Summary

Lithium is the practical polarized nucleus for the EIC because three things coincide: an alkali atom that a single laser polarizes and an existing EBIS charge-breeds; two stable isotopes whose nuclear spin sits on a light cluster — a polarized deuteron in ^{6}Li, a polarized proton in ^{7}Li — bound to an inert α that exact few-body theory describes; and ions whose spin dynamics in the ring are deuteron-like. Its physics is inaccessible without polarization, since b_1, Δ, g_1^A and the tagged tensor asymmetries all vanish or are undefined without it, and largely inaccessible at a fixed target, where dilution, spin-temperature-limited tensor polarization and the loss of slow spectators are structural. Four measurements follow, each a first: b_1 of a nucleus beyond the deuteron, the never-measured double-helicity-flip structure function Δ, the polarized EMC effect on a bound proton at collider kinematics, and α-tagged DIS on a polarized cluster. The requirements they place on the machine are specific — tensor polarization delivered as m = ±1-rich and m = 0-rich bunches alternating bunch by bunch, transverse polarization at the interaction point, relative luminosity between the three b_1 fills controlled to 10^{-4} (the Δ ratio tolerates 10^{-3}; Report 3 [26] Table 9), and a near-beam tagging capability for A/Z = 2 recoils that no published optics provides — and each is a concrete question to the accelerator and detector groups. The projections in the companion report [5] quantify what answering them buys.

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Appendix A · Revision history

DATE	REVISION

2026-09-02	Consistency review of Reports 0–4 (`docs/consistency_review_2026-09-02.md`): 32 findings here, none of them overturning a result. Five printed numbers are re-read from the scripts that produce them — §5.2's R1998 shift is +18, +18, +8 and −4% (F190) and its Δ amplitude 21–43σ per bin, as in Table 3 (F163); §5.3's mean-field valence depletion at x = 0.45 is 0.043 (F160); the second pair of Δ bins in §6 sits at y ≈ 0.025 (F033); and Table 3's reconstructed δÂ is $(0.9–3.0)\times10^{-4}$, on the PYTHIA chain the purities beside it are read on (F031). The remainder are the labels a reader needs in order to act on a number. §4.2 attributes the b_1 signs it displays to this programme's own transcription of [27] rather than to [27], whose convention is the one HERMES and Cosyn et al. apply, and scopes the consequences of the flip to this report (F002, F021, F153). §5.2's coherent reach is labelled as the counting floor of its best (x, Q^2) super-bin, with the seven-|t|-bin fit of the whole tagged sample quoted beside it at 0.07–0.12% per unit P_{zz} per year (F007); its 5 × 41 aperture stands on the decided ePIC baseline proton lattice, adopted for a ^{6}Li fill as an educated guess and worth ×0.64 in the Z/A = 0.5 alternative (F035); and the pot edge is given per axis — horizontal 2.50 / 1.51 / 0.53 mrad against vertical shut / 2.12 / 0.92 mrad — because the sentence applies it per axis and the measured aperture is not square (F122). §2.3 states the transverse axis at IP6 as expected rather than established, an assumption Report 3 §7 now carries with its owner, no rotator or snake configuration for a lithium fill being defined (F137, F172). §5.2's Δ scenario is given an x-moment one third of the bag value, the counting fraction of the polarized pair among six nucleons (F003); §5.3's flavour-scheme effect is attributed to the two grids Table 3 names (F025); §5.4's near-beam tail clears the envelope in 0.20 / 0.01 / 0.08 points, a ninth of the tag at 5 × 40.8 and a thirtieth or less at the other two (F048); and §7's 10^{-4} relative luminosity is the three-fill b_1 ask, the Δ ratio tolerating 10^{-3} (F008). Table 3 gains the conventions its entries are quoted in — $P_{zzz} \approx 0.11$ is the normalization of [20] and 0.34 in the rank-3 one `polligen.spin` adopts, open item 1 of `docs/note_7li_theory_questions.md`, as §2.1's P_{zz} = 0.69 is the convention of [10] (F026) — the toy-input δΔR the target-mass term is weighted on (F022, F158), the tensor-polarimetry scale as the ≤ 5% this programme specifies with 3% the optimistic case (F004), the published seven-bin window 0.017–0.25 GeV² and the |t| > 0.04 GeV² the reference tag selects (F038), and α + d rejection by the |t| shape and the partner-fragment hit as the binding requirement rather than pot charge identification (F139); its caption and the footer say which rows are published here and which are drawn from Report 3, over which ^{7}Li configurations the tags span, and on which fitted unpolarized backbones the signal scenarios ride (F044, F149). The spectator results and the cluster model are cited to Report 3 [26] rather than to [5] (F028, F154), and Figure 2's caption names the panel's "legacy, 275 GeV p" label (F155) and reads the coherent fraction as halving at x_{coh} = 0.01, the coherence length reaching the ^{6}Li diameter only at x ≈ 0.02 rather than at 0.01 (F019, F169). Two rows below are corrected in place: 2026-08-29 says which separations §5.3 carries and which significances Table 3 does (F018), and 2026-08-28 attributes the pot aperture and the over-rigid destination to the Report 3 sections that measure them (F053). No figure is regenerated. Wording suggestions of the same review are applied with them — F016, F017, F023, F024, F027, F029, F032, F039, F040, F046, F047, F052, F147, F148, F150, F151, F152, F156, F157, F159 — which add the two D-state probabilities behind §2.2's 0.811 and their distance from the VMC overlap; the 0.45% by which the deuteron sits above ^{6}Li at the γ-matched points; the $O(\gamma^2)$ identification of the beam and photon axes in §4.2 and, in Figure 1, the two extremes the panels actually draw; in §5.2 the applied-cut coherent triple cited to Report 3 [26] alone, the a_2 footing on which its 8–11σ is quoted, and the second interaction region's ≈ 20% as our interpolation between published efficiencies; in §5.4 the 31 / 22 / 29% as the cluster-model routing against the 2.5% and 25% the S + D generator returns, the BeAGLE factors as the spectator peak's, and Table 2's acceptance hole as the top configuration's; §6's one placeholder luminosity at every configuration; in Table 3 the P_{zz} footing of each amplitude, the b_1 significances as read in each error bin rather than across bins, and the twist-3 residual bin by bin (77 / 13 / 9.4 / 6.8 under the term); Figure 3's abscissa, its right panel's Q^2 slice and its ten-year series; and, in the rows below, the superseded target-mass clause put in the past with the values that moved under it (0.44 / 0.71 / 1.06 → 0.46 / 0.73 / 1.08%) and R1998 named behind its `r_func` hook.
2026-08-29	Verification pass over the round. Table 3's reconstruction row restated on the 2026-08-28 toolchain review — purities 0.56–0.75 and the reconstructed-bin amplitude within 7% of the true-bin value, against the 0.52–0.76 and 5% it had kept from before it. The target-mass clause named which δΔR it was measured against (8.7–40× the toy-input statistics the bound was computed with, 8.5–36× the grid-input δΔR the reach column leads with) — a naming the same day's restatement below removes. The b_1 transfer is given in full — the per-deuteron→per-nucleon ½, the rank-2 0.92 and the ⅓ dilution, 0.15 in all — since 0.31 is the factor applied to a b_1 already per nucleon. §5.3 names the Cloët–Bentz–Thomas curve the ratios are read off (their Eq. 23). §5.4 and Table 3 carry the 5 × 40.8 caveat on the ^{7}Li α: the 97% is an angle-only routing there, and the measured per-band insertion leaves 57%, station 2 only. Table 2's caption says the triton gun was fired at R = 1.286 against the table's physical-mass 1.2897. The footer names Reports 1–4, and this appendix now runs newest first as Reports 2–4 do. Three changes later the same day move numbers here. The unpolarized baseline both polarized-EMC camps are transferred onto is now measured rather than modelled — EPPS21's ^{6}Li over CT18ANLO — EPPS21's own proton baseline, so the proton fit cancels — valence depletion 0.0311 against the mean-field model's 0.0584, which halves the transfer and halves the reach: §5.3's separations read 0.023 / 0.021 / 0.018 / 0.032 and Table 3's g_1^A row 0.55 / 0.53 / 0.31 / 0.17σ per bin at 10 fb^{-1}/u and 1.75 / 1.66 / 0.97 / 0.53σ at 100, with the best window bin at 0.45σ and 1.43σ, and nothing above 1σ per bin anywhere; the spread between available baselines is now the leading uncertainty on that comparison. The generator carries its target-mass term by default, so Table 3's polarized-EMC systematics cell states the term as computed (0.12–1.08% of ΔR) — the three upper values themselves re-read on the run-15 libraries, 0.44 / 0.71 / 1.06 → 0.46 / 0.73 / 1.08%, the exact-kernel rebuild and its δΔR weights moving the weighted term by at most 0.03 points — and the twist-3 uncertainty on g_2 as what is left open (≤ 0.16%), where it had stated the term as a live bias; §4.2 follows. The b_1 signal is read at each error bin's own ⟨Q^2⟩, so Table 3's b_1 row is 2.2×10^{-4} and 1.6 / 4.7 / 7.3 / 10.4σ. And the

	per-configuration far-forward transport became the baseline for the tagging optics, taking §5.4's and Table 3's ^{6}Li α tag to 31 / 22 / 29%. §2.2 states the vector polarization the simulation adopts for ^{6}Li, 0.811 of the nuclear spin per valence nucleon, settled that day on the cluster picture and computed from the same wave function as the tagged α–d observables rather than from a transcribed constant, with 0.81–0.85 carried as the band against the ab-initio VMC reading of the same quantity; it is rank 1 and leaves the b_1 transfer of Table 3 untouched, and no ^{7}Li number depends on it. The tensor-sector sign was settled the same day: the simulation now carries the HERMES and Cosyn et al. convention, $A_{zz} = -(2/3)\, b_1/F_1$ for a spin axis along the photon, in place of the transcription §4.2 writes the master formula in. It is one constant, it affects only the sign of A_{zz} relative to b_1, and no magnitude, error or Δ in this report moves with it.
2026-08-28	The run-plan share stated where §6 quotes the reaches: the observables want different spin states, different far-forward optics — the coherent channel of §5.2 the tagging optics with its luminosity penalty — and, between the polarized EMC effect and the tensor sector, different isotopes, ^{6}Li and ^{7}Li being separate fills and separate runs, each reach being quoted at the whole year in its own, so Table 3's reaches are per observable and cannot be added into one year; how a year is actually divided is a programme decision priced in plans/07 WP2, under which every statistical error grows as one over the square root of the share and a luminosity quoted as a reach does not move. No number here moves.
2026-08-28	The target-mass (γ^2) term that $A_\parallel \approx D \cdot g_1/F_1$ drops implemented behind `InclusiveKernel(target_mass=True)`, default off, and bounded rather than left implicit: §4.2 names the factor $1 + \gamma^2$ the ≈ hides ($\gamma^2 \le 0.010$ rate-weighted over the window of Table 3), and Table 3's polarized-EMC row the bias it leaves behind — ΔR high by 0.12 / 0.44 / 0.71 / 1.06% at x = 0.09 / 0.28 / 0.45 / 0.71, 8.7–40× below the toy-input δΔR of the same weights (8.5–36× against the grid-input one) and 2.8–25× below the polarimetry scale (`evgen/scripts/target_mass_bound.py`). No published number moves.
2026-08-28	The theory curves of §§5.1 and 5.3 and of Table 3 replaced by the published figures themselves, digitized from the papers' vector paths: the polarized and unpolarized EMC ratios of [21] and [53] and the deuteron b_1 of [45] and [46]. The polarized-EMC discrimination stays in the valence region but halves, to 0.84σ per bin at 10 fb^{-1}/u and 2.7σ at 100 fb^{-1}/u in the best bin of the window in which the two calculations are comparable at all, and the b_1 scenarios separate into one that is measurable and one that is not. The ^{6}Li b_1 transfer is the rank-2 tensor dilution ⅓ × 0.92 rather than the rank-1 vector 0.87. The Cosyn–Weiss closed-form tagged tensor asymmetry (§5.4) is added as a validated limit of the generator.
2026-08-28	§5.4 and Table 3 restated on the per-configuration far-forward optics, replacing the retired single proton-derived 73 / 164 μrad pair on which the ^{6}Li α tag was published as 1.9% / 1.5%. §5.2's coherent acceptances are labelled as the circular form of the formula they follow, with the anisotropic envelope and the measured pot edge given beside them. Later the same day the pot aperture and the over-rigid destination were measured in the current ePIC geometry (the aperture: Report 3 §6.1 and Report 4 §6; the destination: Report 3 §5 and Table 6; both measured in plans/09 B1): the pot edge is 2.50 / 1.51 / 0.53 mrad against 2.0 / 1.35 / 1.03, so §5.2's coherent tag reads 9×10^{-10} / 6×10^{-27} / 7×10^{-14} and the binding constraint is the silicon at 5 × 40.8 and the beam at 18 × 137.5; the ^{6}Li α tag is 1.9 / 1.7 / 2.6% at the Yellow Report divergences and 28–35% under the tagging optics; and the ^{7}Li triton, routed as lost while R = 1.29 had no window, is measured onto the Roman-Pot silicon in 60 of 60 events with a calorimeter deposit behind it and is tagged in 78 / 92 / 94%, which opens the α + t double tag.
2026-08-27	Rewritten as a paper: abstract, a single logical thread (atom → nucleus → ion; polarization → collider; formalism → observables → requirements), references numbered in citation order, and every repeated fact stated once. All numbers restated from the current scripts at the γ-matched beam energies (^{6}Li at 40.8 / 99.5 / 137.5 GeV/u, plans/10): the sweet spots, amplitudes and errors of the Δ projection, the R-model shifts, the b_1 and polarized-EMC reach, and the coherent-channel acceptance, which with the Yellow Report divergences does not survive at IP6 with any published optics. The pre-2026-08-27 coherent and tagging results computed at the superseded rigidity-scaled energies (the measured-aperture survival at 20.5 GeV/u, the ^{6}Li α tag at 50 GeV/u) are withdrawn; the 13.5% of a 0.20 GeV envelope is retained only as the reference tag of the projections, no longer as an IP6 optics.
2026-08-26	Development run 8 folded in: the PYTHIA 8 hadronic final state, the Roman-Pot aperture measured in the ePIC geometry, the published R1998 fit behind an `r_func` hook, physical nuclear masses in the far-forward routing.
2026-08-25	References verified against the local copies in refs/ (two citations corrected); the spin-state ratio and bunch-by-bunch alternation as the Δ requirement; the two-state error formula.
2026-08-21	The why-lithium, why-polarization and fixed-target-versus-collider material (now §§2–3) rewritten against primary sources, and the requirements table added.
2026-08-17	First version: the DIS and spin-1 formalism from first principles, and the four flagship measurements (b_1, Δ, the polarized EMC effect, spectator-tagged DIS) with their reach.

Data availability: the code, figures and reproduction instructions are in the PolarizedLithiumSim repository (`docs/reproduction_manual.md`); companion documents are Reports 1–4 of this series. Projections use scenario signal structure functions with explicit bands, on fitted unpolarized backbones (CT18NLO, NNPDFpol11 and, for the polarized-EMC baseline of §5.3, EPPS21).